# Large Language Models (LLMs)

## 1. What is an LLM?

A Large Language Model is a machine learning model designed to process and generate natural language. LLMs learn statistical patterns from large collections of text and use those patterns to predict suitable sequences of tokens. They can be used for question answering, summarization, translation, information extraction, dialogue, and code generation.

## 2. Tokens and Context

LLMs process text as tokens rather than directly as complete sentences. A token can represent a word, part of a word, punctuation, or another piece of text. The model receives a sequence of tokens within a context window and uses that context when producing its response.

## 3. Transformer Architecture

Modern LLMs commonly use the Transformer architecture. Transformers use attention mechanisms to model relationships between tokens. Self-attention allows the model to consider different parts of an input sequence when interpreting each token.

## 4. Generation

During generation, an LLM predicts the next token based on the preceding context. The process is repeated until the response is complete or a stopping condition is reached. Generation settings can influence how deterministic or varied the output is.

## 5. Applications

LLMs can support conversational assistants, document analysis, coding assistants, summarization systems, search interfaces, and agentic applications. In these systems, the LLM often acts as the language and reasoning component while external tools provide information or actions.

## 6. Limitations

An LLM can produce incorrect or unsupported information. Its pretrained knowledge may also be incomplete or outdated. For applications requiring specific or current information, systems can provide the model with external context through retrieval, tools, databases, or APIs.